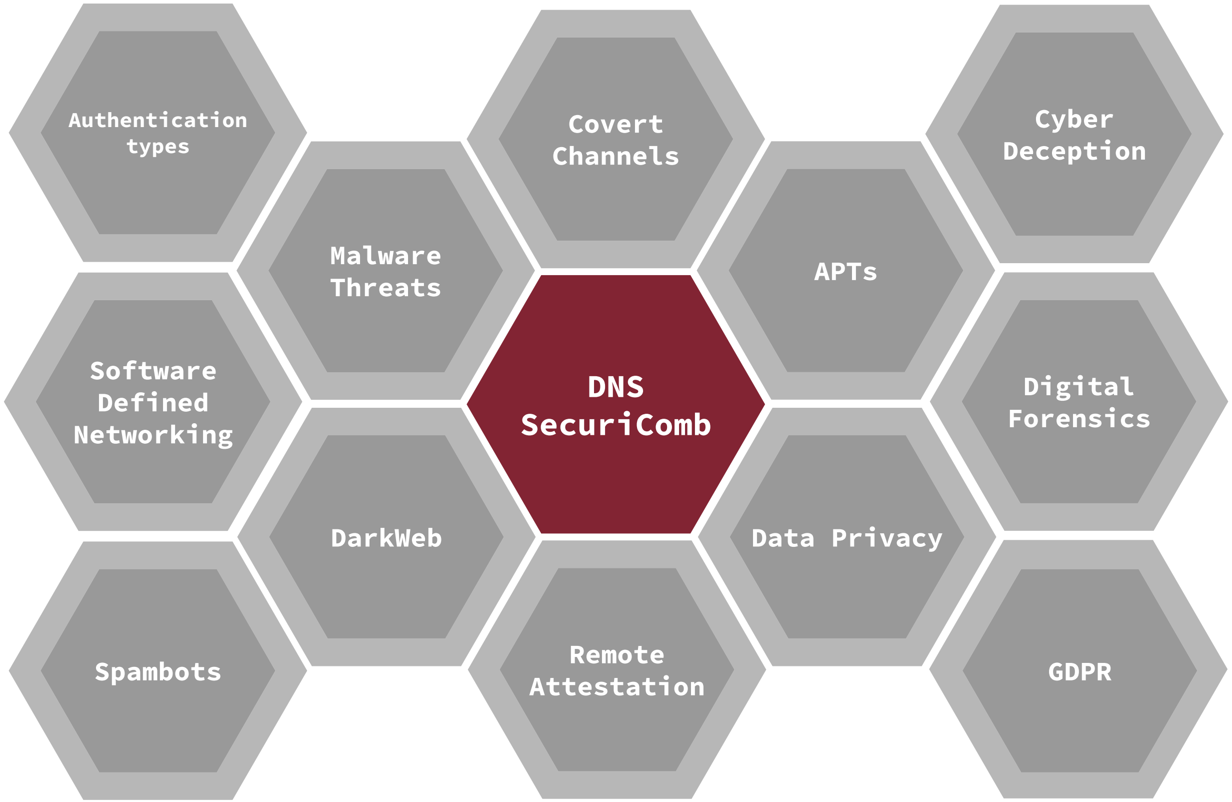
Data and Network Security

(Master Degree in Computer Science and Cybersecurity)

Lecture 3
Spring 2026

Sapienza Università di Roma

SecuriComb


Authentication types
Malware Threats
Covert Channels
APTs
Cyber Deception
Software Defined Networking
DNS SecuriComb
Digital Forensics
DarkWeb
Data Privacy
Spambots
Remote Attestation
GDPR

SecuriComb

Authentication types

Covert Channels

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Malware Threats

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GDPR

Outline for today

- Recap
- How malware is spread?
- Emerging trends

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- **Recap**
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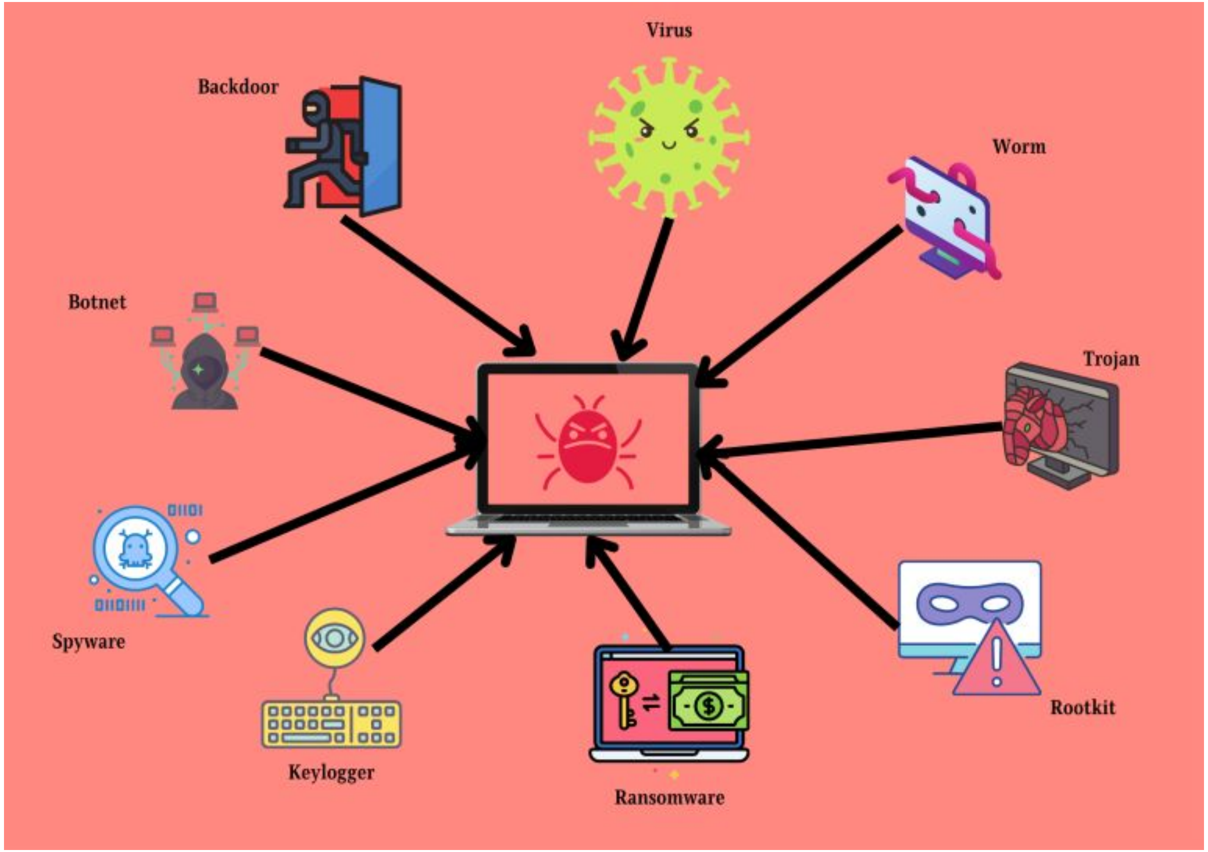
- Emerging trends

Malware Types



Ransomware

Type of malware from cryptovirology that threatens to publish the victim's data or perpetually block access to it unless a ransom is paid.

Signature based vs. Behaviour based detection

Outline for today

- Recap
- **How malware is spread?**
- Emerging trends

How malware is spread?

- Email attachments
- Internet downloads
- File sharing networks
- Removable media
- …

How malware is spread?

- **Phishing emails:** Attackers send fraudulent emails with malicious links or attachments, and the users, by clicking on links (or opening malicious attachments), unknowingly download/execute malware.

How malware is spread?

- USB and other removable media: malware is already prepared and stored in the removable media (usually autorun scripts)

How malware is spread?

- **Drive-by-downloads:** When you visit a compromised (or malicious) site, malware is downloaded automatically in your system. A large majority of such malware relies on browser vulnerabilities.

How malware is spread?

- **Malvertising:** Malware is embedded in online ads and clicking on those ads the users are redirected to the malicious site and we end up to the drive-by-download scenario

How malware is spread?

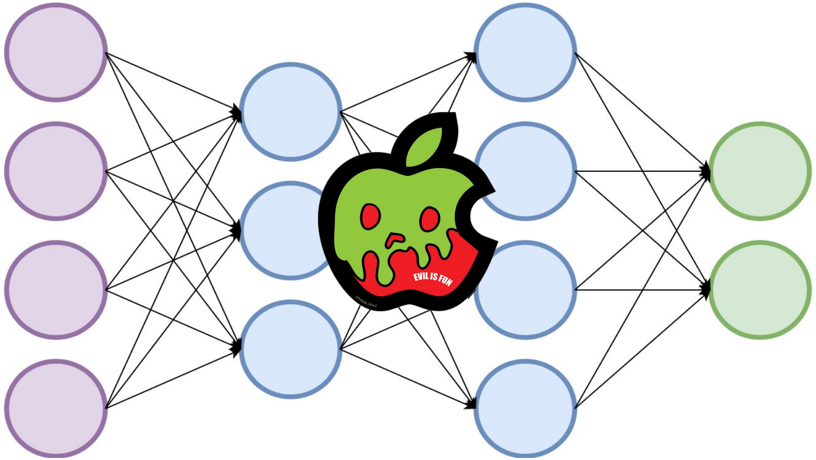
- **File sharing networks:** Malware is embedded in pirated software, movies, or cracked applications shared through P2P networks.

How malware is spread?

- **Software vulnerabilities:** Malware leverages unpatched software flaws to gain access

How malware is spread again?

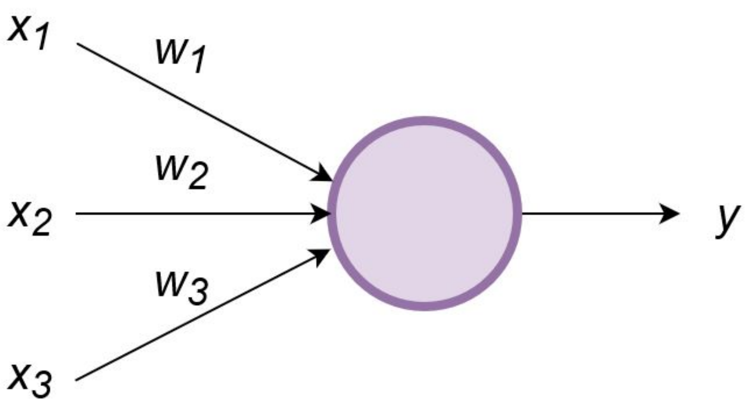
GIST - Malware can also hide inside a deep neural network

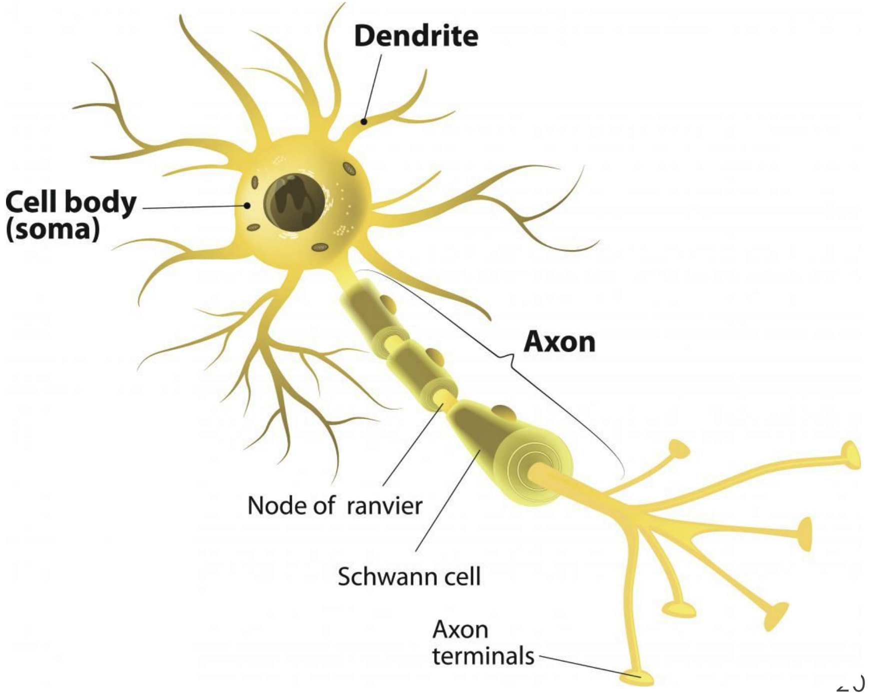


Preliminaries - Neural Network

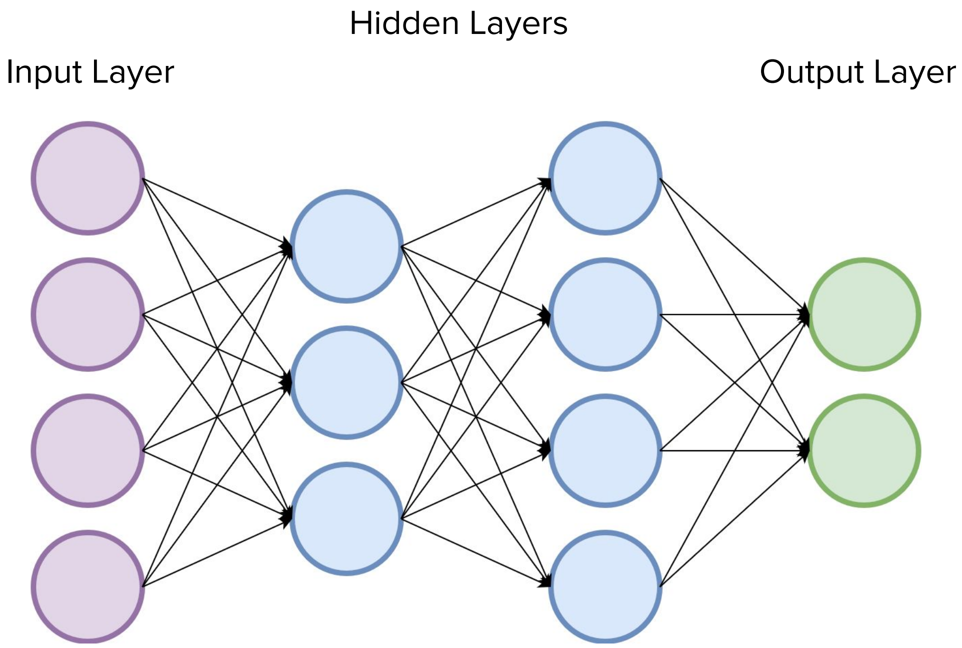
Deep Neural Networks

Why the name Neural Networks?

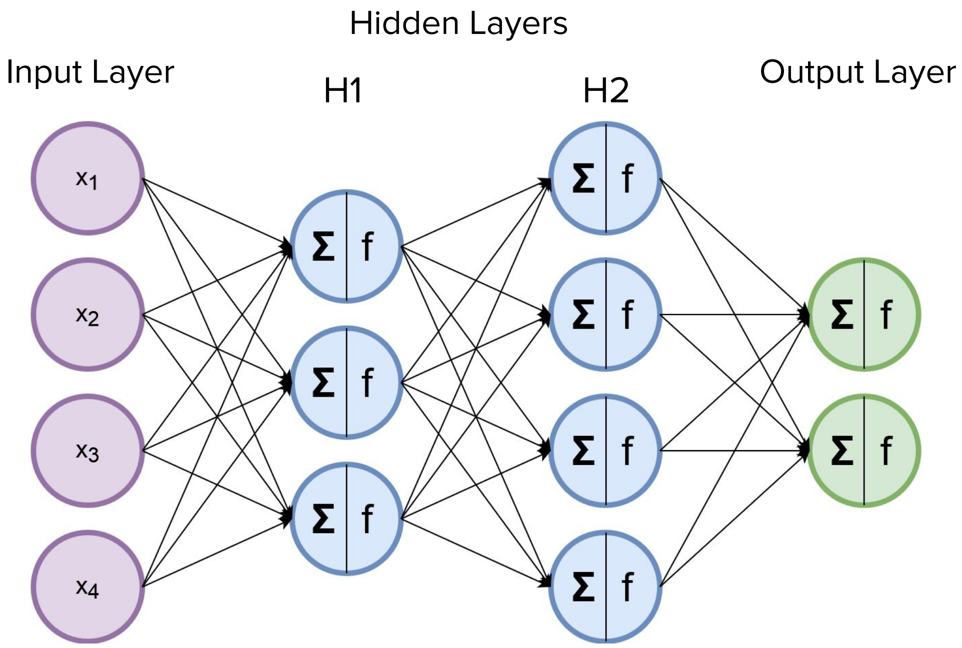




Deep Neural Networks



Deep Neural Networks



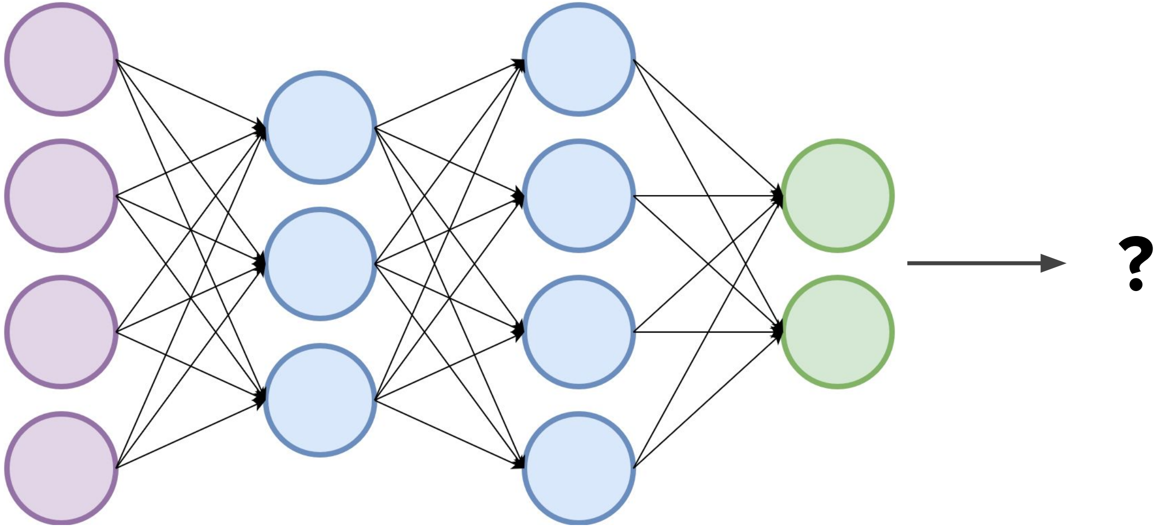
Deep Neural Networks

Ok, got it but ***how*** do they learn?

Deep Neural Networks: Learning

Ok, got it but ***how*** do they learn?

vs



Deep Neural Networks: Learning

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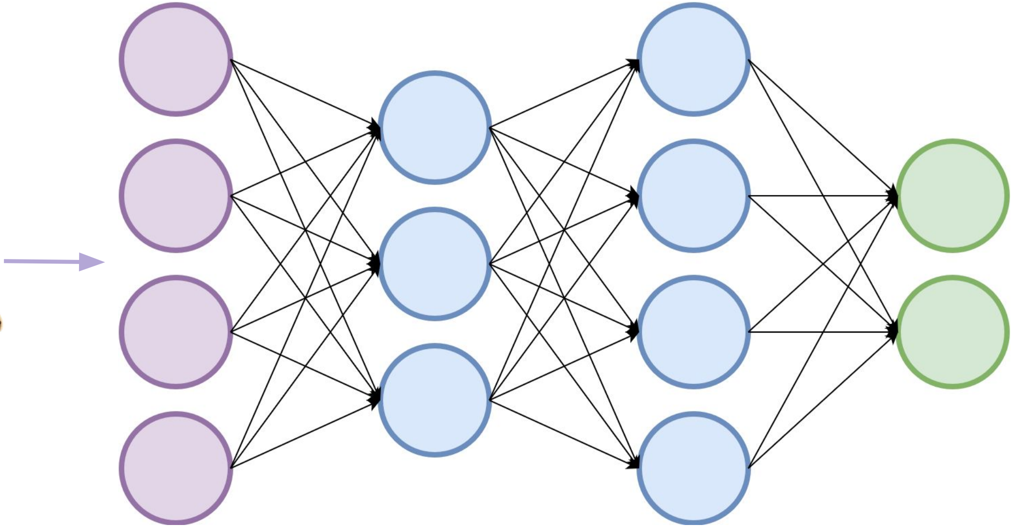

Deep Neural Networks: Learning

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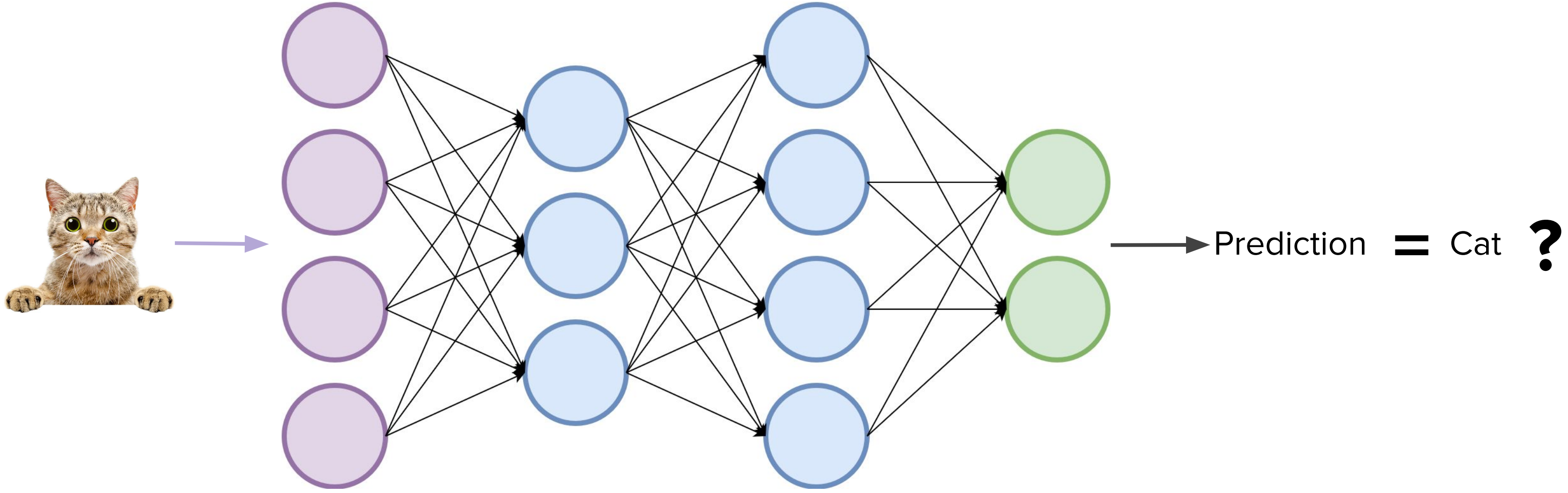
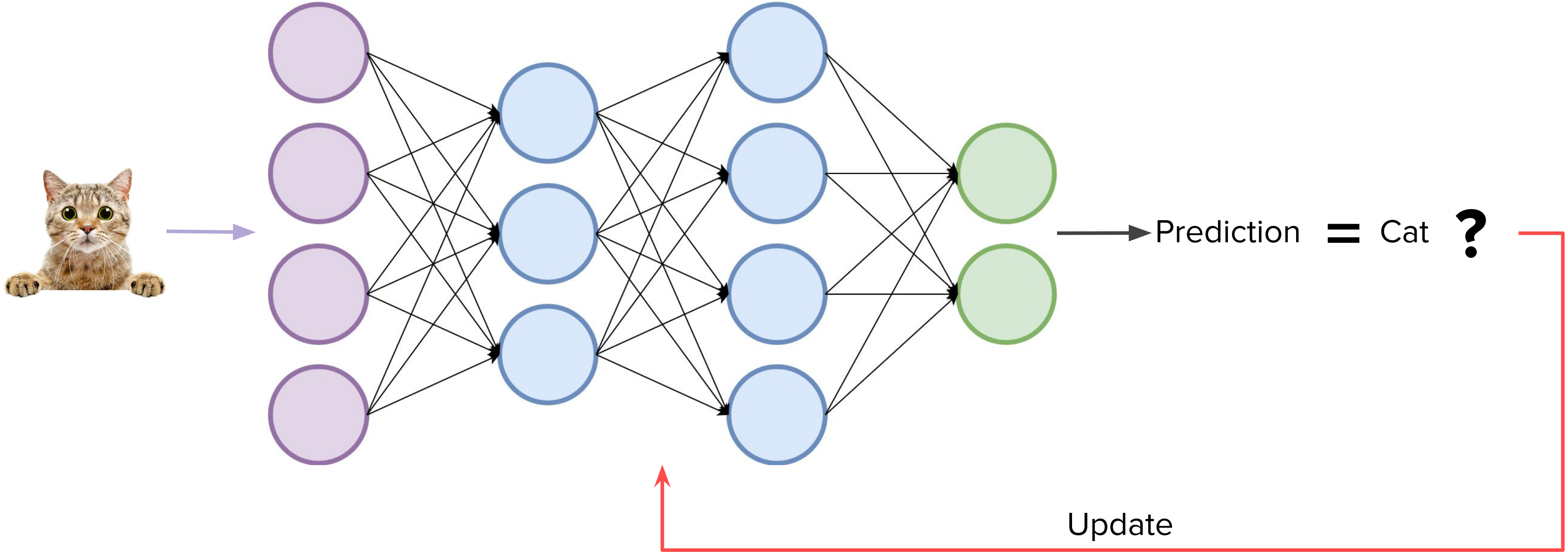


Deep Neural Networks: Learning

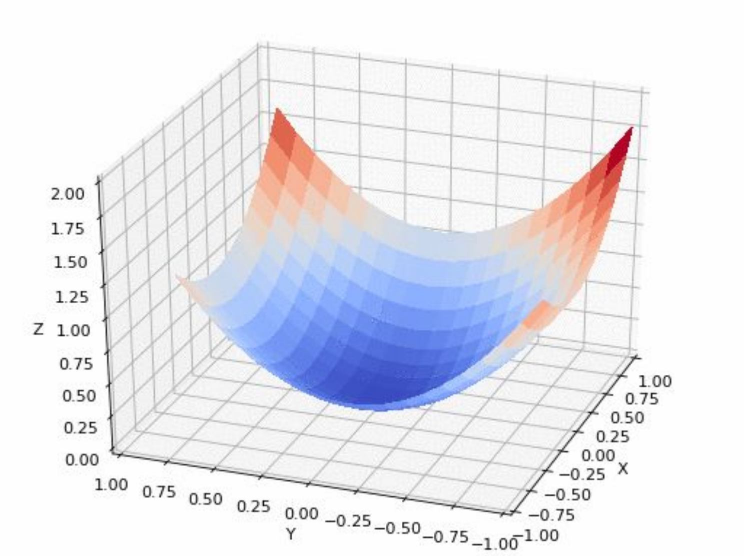
Ok, got it but ***how*** do they learn?

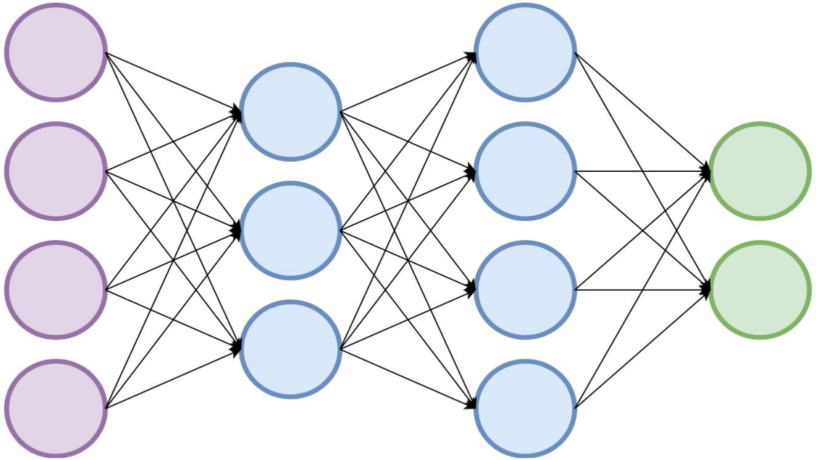


Deep Neural Networks: Learning

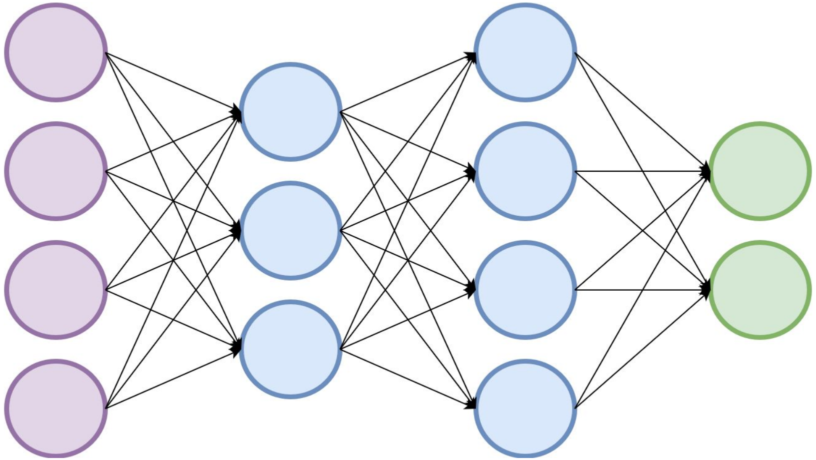
A bit more formally, a DNN defines a *function* to perform a given task

- An error function between the output of the network and the actual output is minimized





A neural network is a computational model inspired by the structure and functioning of the human brain. It consists of interconnected nodes, or "neurons," organized in layers.



Information flows through the network, undergoing transformations at each layer, to perform tasks such as pattern recognition, classification, and prediction.

AI is rapidly becoming instrumental in an ever-growing number of areas

MIT Technology Review

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BIOTECHNOLOGY AND HEALTH

DeepMind is using AI to pinpoint the causes of genetic disease

Fresh from solving the protein structure challenge, Google's deep-learning outfit is moving on to the human genome.

By Antonio Regalado

September 19, 2023

GOOGLE DEEPMIND

The generative AI revolution in the games industry: A path to boundless creativity

October 6, 2023

McKinsey & Company

Operations

Generative AI in finance: Finding the way to faster, deeper insights

February 16, 2024 | McKinsey Direct

RESEARCH

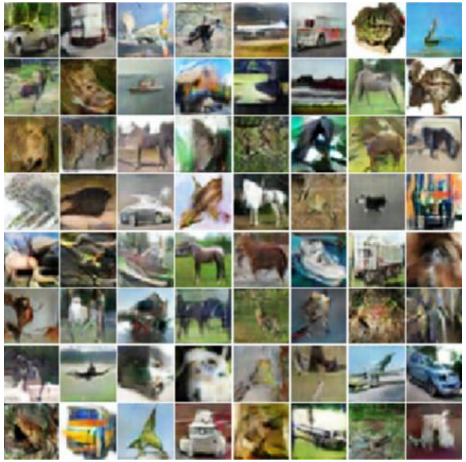
Generative AI Is Set to Shake Up Education

Dec 22, 2023

While educators debate the risks and opportunities of generative AI as a learning tool, some education technology companies are using it to increase revenue and lower costs.

https://www.technologyreview.com/2023/09/19/1079871/deepmind-alphamissense-ai-pinpoint-causes-genetic-disease/
https://cloud.google.com/blog/products/gaming/generative-ai-in-the-games-industry
https://www.mckinsey.com/capabilities/operations/our-insights/generative-ai-in-finance-finding-the-way-to-faster-deeper-insights#/
https://www.morganstanley.com/ideas/generative-ai-education-outlook


MNIST images


CIFAR-10 images


faces

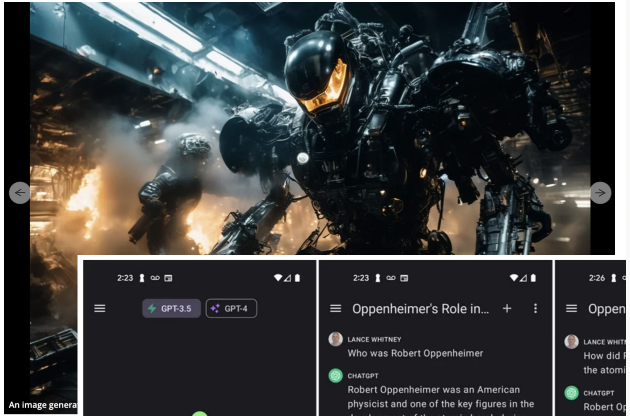

album covers

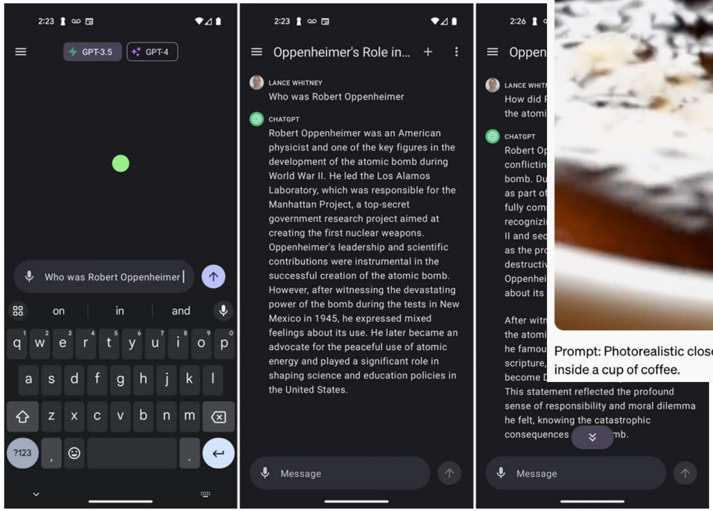
In less than a decade, we went from this...


bedrooms

images from:
- https://blog.openai.com/generative-models/
- Goodfellow et al. Generative Adversarial Networks
- Radford et al. Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks

To this...





Prompt: Photorealistic closeup video of two pirate ships battling each other as they sail inside a cup of coffee.

0:01 / 0:15

https://openai.com/sora
https://arstechnica.com/information-technology/2023/07/stable-diffusion-xl-puts-ai-generated-visual-worlds-at-your-gpus-command/
https://www.zdnet.com/article/official-chatgpt-app-for-android-finally-launches/

The cost of Building a high-performance DNN

Huge computational power

Large quantities of data

The cost of Building a high-performance DNN

Huge computational power

Large quantities of data

Users prefer to consume the Pre-Trained Model

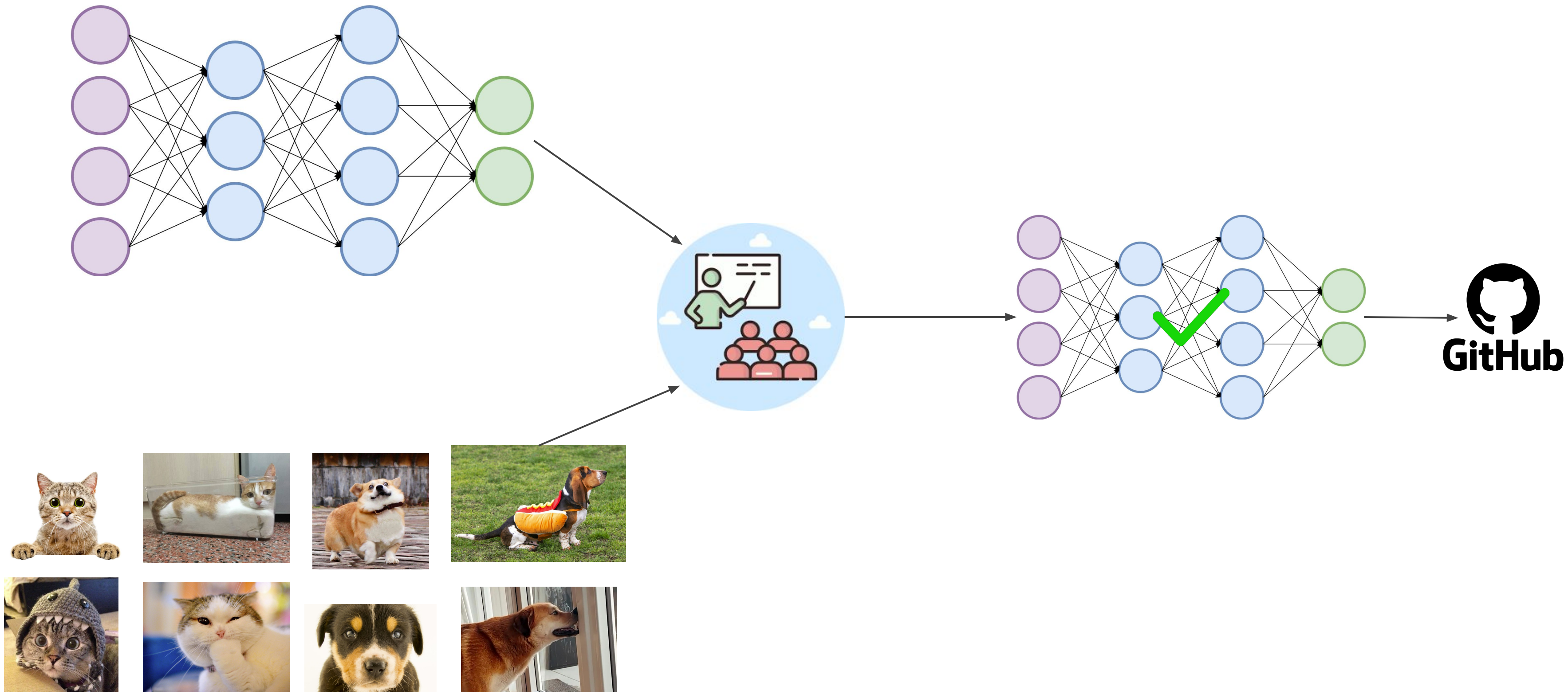
Hugging Face

kaggle

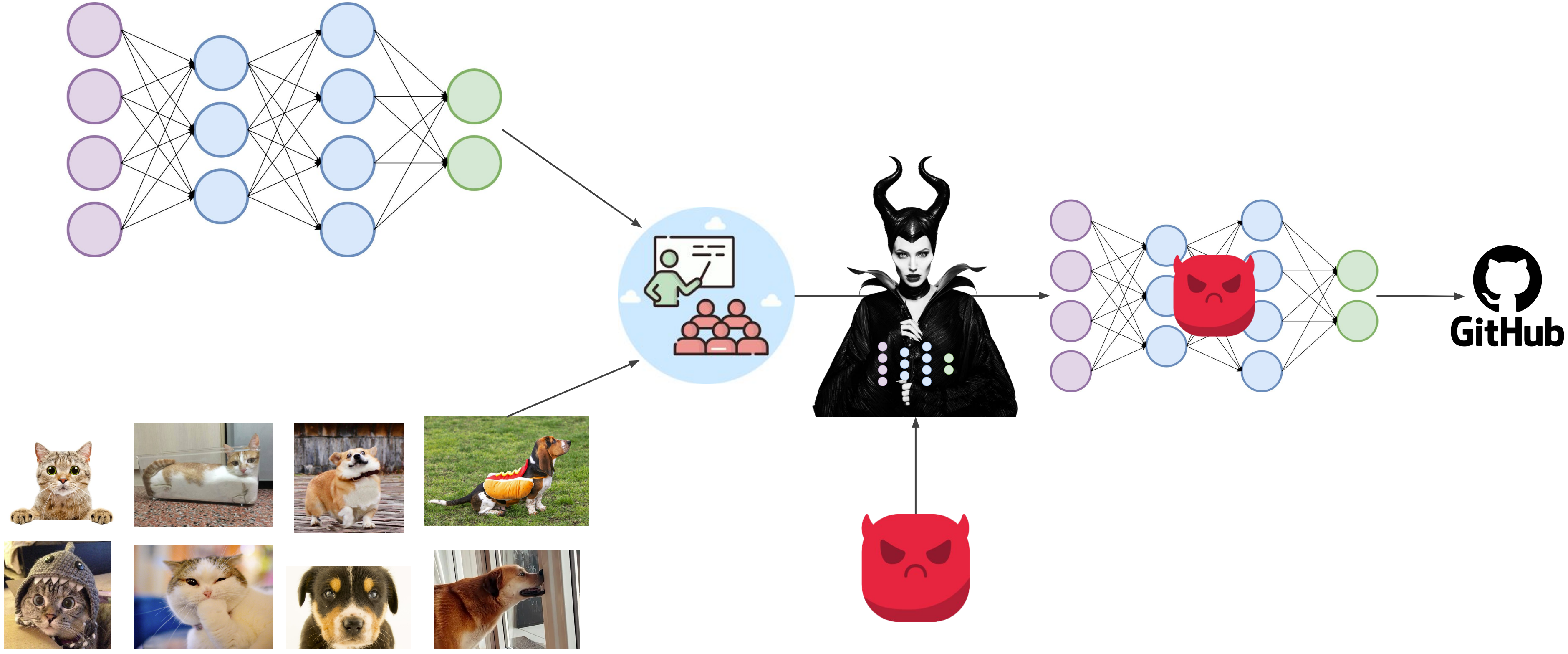
Model Zoo

GitHub

Do you trust your model?



Do you trust your model?



To run these DNNs…

- You need specific software/libraries
  - TensorFlow
  - Pytorch
  - …

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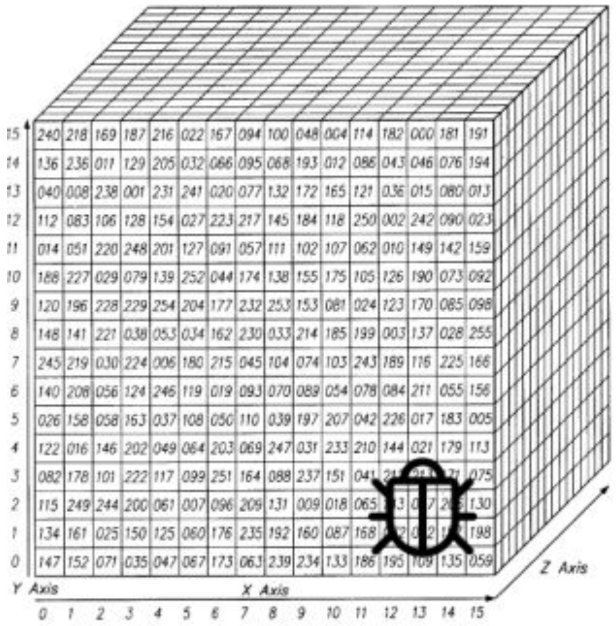
  - TensorFlow
  - Pytorch
  - …

TensorFlow saves variables in binary *checkpoint files* that map variable names to tensor values.

Caution: TensorFlow model files are code. Be careful with untrusted code. See Using TensorFlow Securely for details.

How are these DNNs structured again?

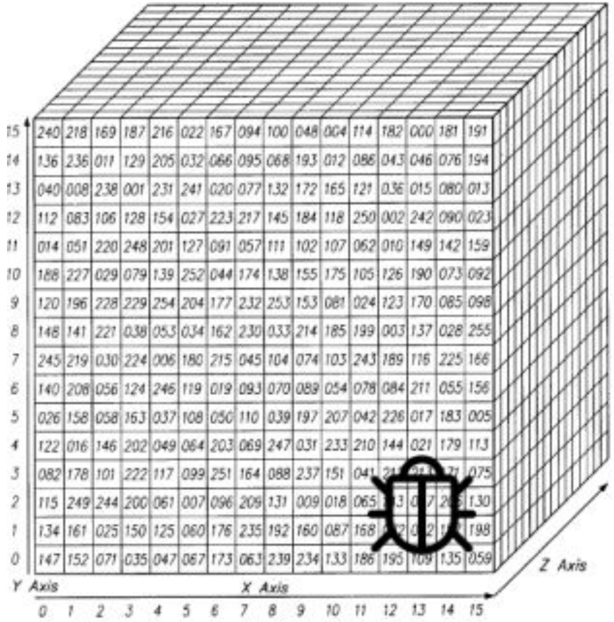
- In the end of days, DNN weights are essentially a matrix of floating point numbers.



How are these DNNs structured again?

- In the end of days, DNN weights are essentially a matrix of floating point numbers.

- These values can be altered by also making sure the performance of the DNN on its intended task remains the same.



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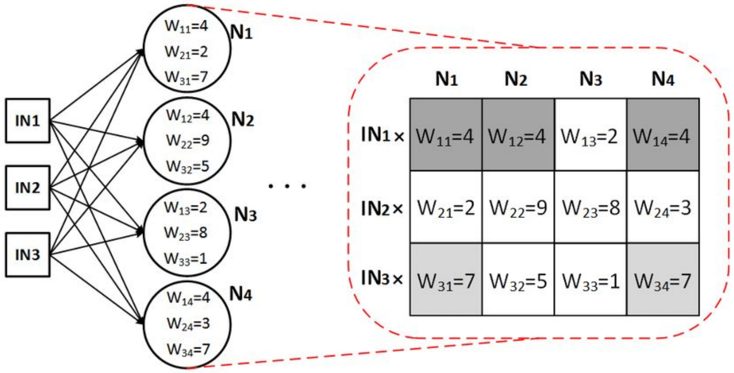
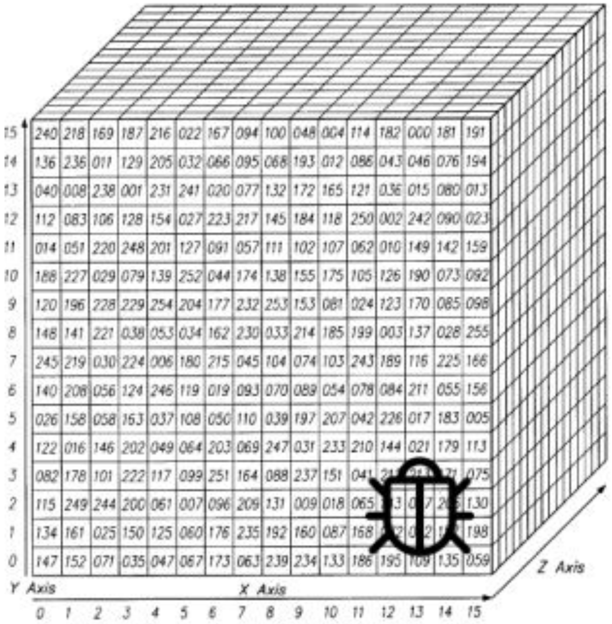




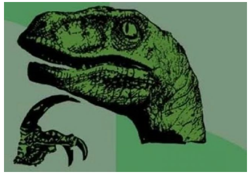
How are these DNNs structured again?

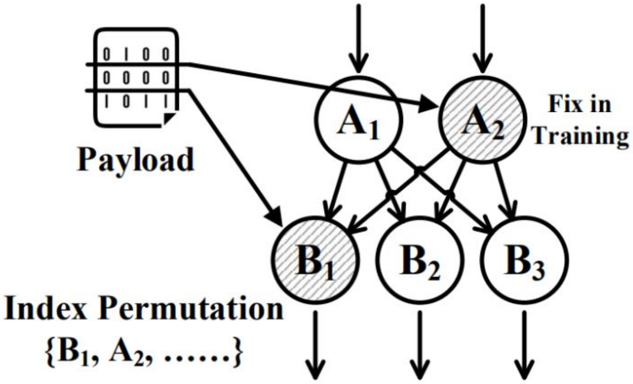
st_dict['model.features.1.weight']

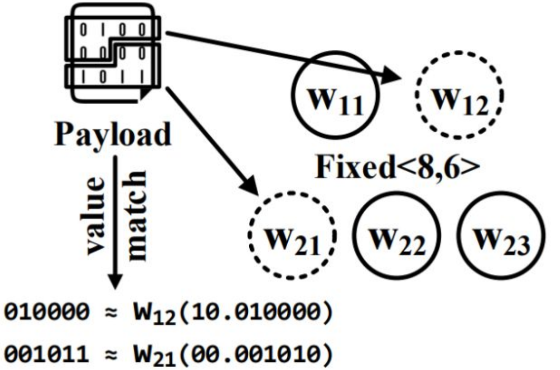
	0
0	0.65384
1	0.00001
2	0.18967
3	0.00000
4	0.47970
5	-0.00000
6	0.48381
7	0.00000
8	0.16715
9	-0.00000
10	0.50307
11	0.35800
12	0.67103
13	0.32597
14	0.27947
15	0.62527
16	-0.00000
17	0.13474
18	0.53121
19	0.19264
20	0.19160
21	0.48617
22	0.31987
23	0.05421
24	0.45253

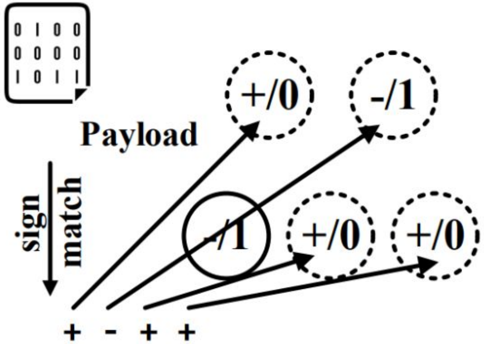
```
OrderedDict([('model.features.0.weight', tensor([[[[-5.9550e-01, -8.3957e-01, -5.1592e-01],
          [ 4.3290e-02, -5.4894e-02, -1.9865e-01],
          [ 6.0978e-01,  9.1177e-01,  6.1916e-01]],
         [[-9.0904e-01, -1.3191e+00, -8.8276e-01],
          [ 7.5378e-02,  6.8023e-03, -1.9233e-01],
          [ 8.9835e-01,  1.3841e+00,  9.1575e-01]],
         [[-3.8599e-01, -6.1602e-01, -3.4967e-01],
          [ 1.3486e-02,  3.0266e-02, -1.0543e-01],
          [ 3.4950e-01,  6.8686e-01,  3.9954e-01]]],
        [[[ 1.0105e-05, -5.7597e-07,  3.7357e-05],
          [ 2.3276e-05, -2.6341e-05,  2.4588e-05],.................
```

How to embed a malicious payload into a DNN?

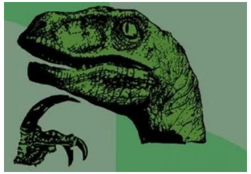


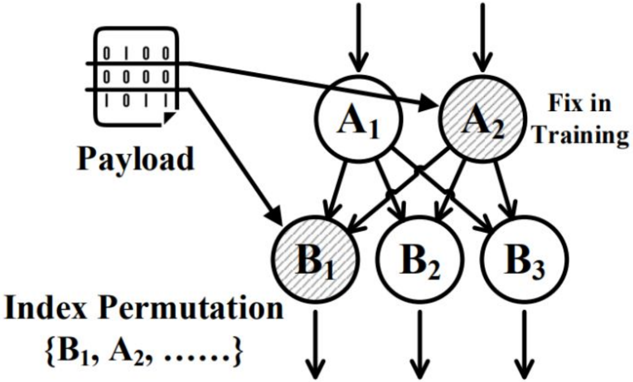






Resilience Training

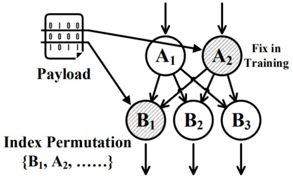




Resilience Training: Fix some neurons (parameter) and train the network to perform good on the intended legitimate task without altering those nodes.

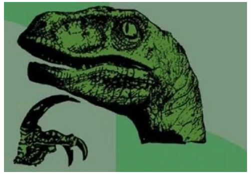
- Malware bits are spread in the fixed parameters
- Need to collect in the appropriate order to have the correct malware binary
- Detectable?
- Robust?

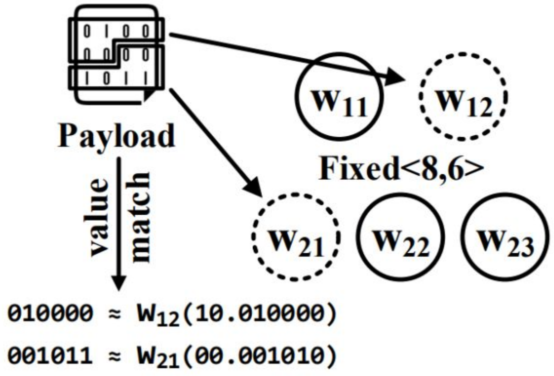
Resilience Training (Model impact)



		Original	EquationDrug	ZeusVM-decypted	Cerber	Ardamax	NSIS	Kelihos	Mamba	WannaCry
Medium DNNs	Resnet50	75.2%	75.2%	75.4%	74.8%	74.7%	75.1%	75.3%	74.6%	74.8%
	Inceptionv3	78%	78.3%	78.4%	78.2%	77.9%	78.4%	78.1%	77.6%	78.4%
	Densnet201	77%	77.2%	76.7%	77.1%	76.9%	77.3%	76.5%	77.2%	77.3%
	Resnet18	70.7%	71.1%	71.2%	70.9%	71%	70.4%	70.3%	70.9%	71.3%
Small DNNs	Googlenet	69.8%	70.3%	69.2%	69.6%	71%	70.5%	69.3%	70.4%	70.2%
	Comp.VGG-16	70.5%	68.3%	69.1%	71.2%	69.1%	68.4%	**63.4%**	**56.1%**	**22.6%**
	Comp.Alexnet	57%	55.4%	56.7%	57.2%	54.1%	**38.2%**	**34.3%**	**16.7%**	**3.9%**
	Squeezenet	57.5%	56.8%	54.3%	53.2%	**48.3%**	**35.4%**	**29.6%**	**15.1%**	**4.1%**
	Mobilenet	70.9%	71.2%	68.5%	66.7%	**54.4%**	**32.5%**	**29.1%**	**6.1%**	**0.7%**

Value Mapping

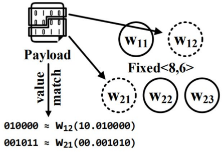




Value mapping: Train the network to perform good on the intended legitimate task without altering anything:

- Malware bits are mapped in some parameters of the DNN
- Need to collect in the appropriate order to have the correct malware binary.
- Detectable?
- Robust?

Value Mapping (Model impact)



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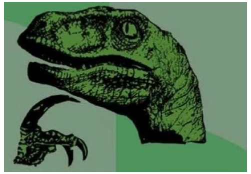
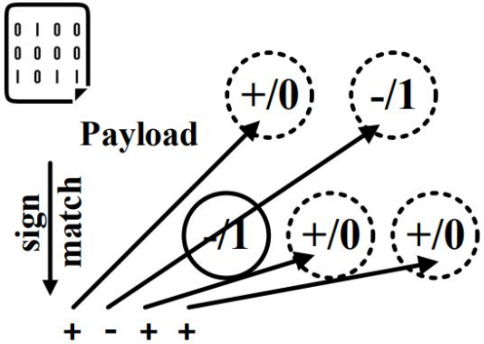
Medium DNNs	Resnet50	75.2%	74.8%	74.7%	75.1%	75.3%	74.6%	74.8%	74.6%	75.7%
	Inceptionv3	78%	78.3%	78.4%	78.2%	78%	77.2%	78.3%	78.4%	78.1%
	Densnet201	77%	77.1%	76.9%	77.3%	76.5%	77.2%	76.5%	77.3%	75.9%
	Resnet18	70.7%	71.1%	70.2%	70.5%	71.6%	72.1%	70.9%	71%	70.4%
Small DNNs	Googlenet	69.8%	70.1%	68.3%	70.2%	68.1%	70.3%	69.8%	68.4%	68.1%
	Comp.VGG-16	70.5%	71.3%	71%	68.3%	69.1%	71.2%	69.1%	69.2%	**48.7%**
	Comp.Alexnet	57%	56.9%	56.7%	57.2%	54.1%	-	-	-	-
	Squeezenet	57.5%	57.3%	56.9%	55.7%	56.8%	**39.7%**	**43.2%**	**21.8%**	-
	Mobilenet	70.9%	69.2%	71%	70.5%	70.3%	**54.7%**	**48.6%**	**49.3%**	-

Sign Mapping

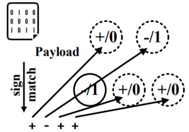




Sign mapping: Train the network to perform good on the intended legitimate task without altering anything:

- Malware bits are mapped in some parameters of the DNN (following the sign of the parameter)
- Need to collect in the appropriate order to have the correct malware binary.
- Detectable?
- Robust?

Sign Mapping (Model Impact)



		Original	EquationDrug	ZeusVM-decypted	Cerber	Ardamax	NSIS	Kelihos	Mamba	WannaCry
Large DNNs	Vgg19 (17.16MB)	71.1%	71.3%	71.2%	70.7%	71.2%	70.9%	71.1%	71.2%	71.1%
	Vgg16 (16.45MB)	70.5%	71%	70.6%	69.9%	70.2%	71.1%	70.8%	70.2%	69.8%
	Alexnet (7.27MB)	57%	57.2%	57.1%	56.7%	57.3%	57.2%	56.8%	56.9%	57.3%
	Resnet101 (5.32MB)	77.1%	77.2%	77.4%	76.7%	76.3%	77%	77.3%	77.5%	-
Medium DNNs	Resnet50 (3.05MB)	75.2%	74.8%	75.3%	75.1%	75.5%	75.4%	75.3%	74.9%	-
	Inceptionv3 (2.84MB)	78%	77.4%	78.2%	78.1%	77.8%	78%	78.1%	-	-
	Densnet201 (2.38MB)	77%	77.3%	77.2%	76.4%	76.7%	77.1%	76.4%	-	-
	Resnet18 (1.39MB)	70.7%	71.1%	70.8%	71%	71.2%	69.5%	-	-	-
Small DNNs	Googlenet (0.83MB)	69.8%	68.3%	70.2%	70.1%	-	-	-	-	-
	Comp.VGG-16 (1.34MB)	70.5%	71.3%	71%	70.4%	69.6%	-	-	-	-
	Comp.Alexnet (0.83MB)	57%	55.4%	56.7%	56.3%	-	-	-	-	
	Squeezenet (0.15MB)	57.5%	-	-	-	-	-	-	-	-
	Mobilenet (0.5MB)	70.9%	68.3%	-	-	-	-	-	-	-

Detectable/Robust?

Baselines	Selected Malware Samples					
	Asprox	Bladabindi	Destover	Kovter	Stuxnet	ZeusVM
Vanilla-malware	72.97%	75.68%	83.78%	62.16%	89.19%	91.89%
Stegomalware	8.11%	10.81%	13.51%	5.41%	0.00%	8.11%
*LSB substitution	0.00%	2.70%	2.70%	0.00%	0.00%	0.00%
*Sign-mapping	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

Detectable/Robust?

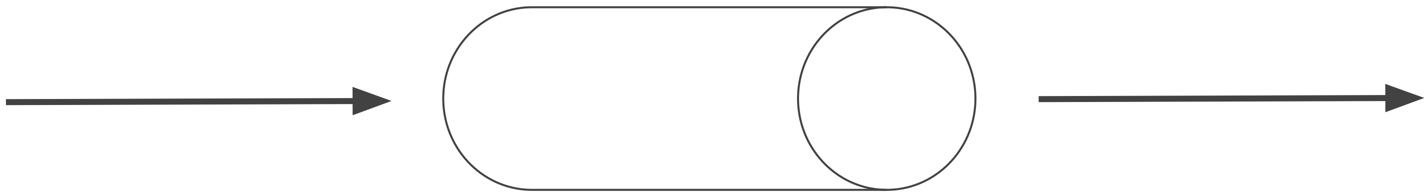
Baselines	Selected Malware Samples					
	Asprox	Bladabindi	Destover	Kovter	Stuxnet	ZeusVM
Vanilla-malware	72.97%	75.68%	83.78%	62.16%	89.19%	91.89%
Stegomalware	8.11%	10.81%	13.51%	5.41%	0.00%	8.11%
*LSB substitution	0.00%	2.70%	2.70%	0.00%	0.00%	0.00%
*Sign-mapping	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

How malware is spread again?

Preliminaries - Digital (stealthy) communication

Digital Communication

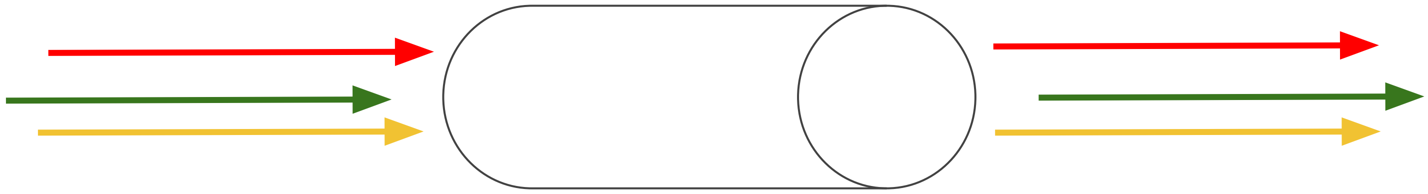
- Consists of the transfer of data or information using digital signals over a point-to-point channel.
- Data or information are digitally encoded as discrete signals, and are transferred through a communication channel.



Communication channel

What is CDMA?

- Channel coding technique by which a narrowband signal is spread into a wider bandwidth.
- Codes the data at a higher frequency by using pseudorandomly generated codes that the receiver knows.
- developed in the 1950s for military communications:
  - resist the enemy efforts to jam the communication channel.
  - hide the fact that the communication is taking place.



Communication channel

CDMA 101

Binary sequence: **[0, 1, 1]**

Assuming ω = 0: ↓

PSK **[-1, 1, 1]**

Using Phase-Shift Keying:

- 0 → -cos(ωt)
- 1 → cos(ωt)
  - ω is the transmitting frequency
  - 0 ≤ t < T

CDMA 101

`Binary sequence:` **`[0, 1, 1]`**

`PSK` **`[-1, 1, 1]`**

`Spreading code:` **`[-1, 1, -1, -1, 1]`**

CDMA 101

Binary sequence: **[0, 1, 1]**

PSK **[-1, 1, 1]**

Spreading code: **[-1, 1, -1, -1, 1]**

Chip sequence: **[1, -1, 1, 1, -1, -1, 1, -1, -1, 1, -1, 1, -1, -1, 1]**

CDMA 101

Binary sequence: **[0, 1, 1]**

PSK **[-1, 1, 1]**

Spreading code: **[-1, 1, -1, -1, 1]**

Chip sequence: **[1, -1, 1, 1, -1, -1, 1, -1, -1, 1, -1, 1, -1, -1, 1]**

-5 **+5** **+5**

CDMA example communication - 2 users

Binary sequence: **[0, 1]**

PSK **[-1, 1]**

Spreading code: **[-1, -1, 1, 1]**

Chip sequence: **[1, 1, -1, -1, -1, -1 ,1 , 1]**

Binary sequence: **[0, 0]**

[-1, -1]

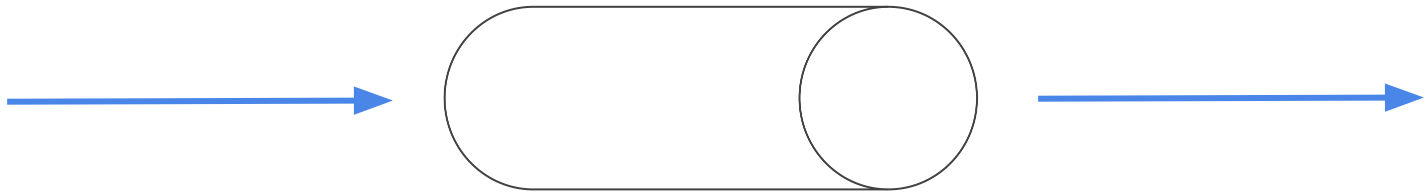
[1, -1, 1, -1]

[-1, 1, -1, 1, -1, 1, -1, 1]

CDMA example in communication

User #1 **[1, 1, -1, -1, -1, -1, 1, 1]** User #2 **[-1, 1, -1, 1, -1, 1, -1, 1]**

Combined signal: [0, 2, -2, 0, -2, 0, 0, 2] - **two users worth of data**



CDMA example - Decode user #1

Combined signal: [0, 2, -2, 0, -2, 0, 0, 2] - two users worth of data

Spreading code: **[-1, -1, 1, 1]**

CDMA example - Decode user #1

Combined signal: **[0, 2, -2, 0, -2, 0, 0, 2] - two users worth of data**

Spreading code: **[-1, -1, 1, 1]**

[-1, -1, 1, 1, -1, -1, 1, 1]

CDMA example - Decode user #1

Combined signal: `[0, 2, -2, 0, -2, 0, 0, 2] - two users worth of data`

✖

`[-1, -1, 1, 1, -1, -1, 1, 1]`

$$\underbrace{[0,\ -2,\ -2,\ 0,}_{\text{Add up}}\ \underbrace{2,\ 0,\ 0,\ 2]}_{\text{Add up}}$$

CDMA example - Decode user #1

Combined signal: `[0, 2, -2, 0, -2, 0, 0, 2] - two users worth of data`

×

`[-1, -1, 1, 1, -1, -1, 1, 1]`

`[0, -2, -2, 0, 2, 0, 0, 2]`

−4 +4

CDMA example - Decode user #1

Combined signal: **[0, 2, -2, 0, -2, 0, 0, 2] - two users worth of data**

[-1, -1, 1, 1, -1, -1, 1, 1]

[0, -2, -2, 0, 2, 0, 0, 2]

-4 +4

What was the length of the spreading code?

CDMA example - Decode user #1

Combined signal: **[0, 2, -2, 0, -2, 0, 0, 2] - two users worth of data**

✖

[-1, -1, 1, 1, -1, -1, 1, 1]

[0, -2, -2, 0, 2, 0, 0, 2]

We have to divide by 4

-4 +4

$\boxed{-1 \quad +1}$

CDMA example - Decode user #1

Combined signal: [0, 2, -2, 0, -2, 0, 0, 2] - two users worth of data

✖

[-1, -1, 1, 1, -1, -1, 1, 1]

[0, -2, -2, 0, 2, 0, 0, 2]

-4 +4

-1 +1

Remember PSK mapping
-1 was 0
and
+1 was 1

CDMA example - Decode user #1

Combined signal: **[0, 2, -2, 0, -2, 0, 0, 2] - two users worth of data**

[-1, -1, 1, 1, -1, -1, 1, 1]

[0, -2, -2, 0, 2, 0, 0, 2]

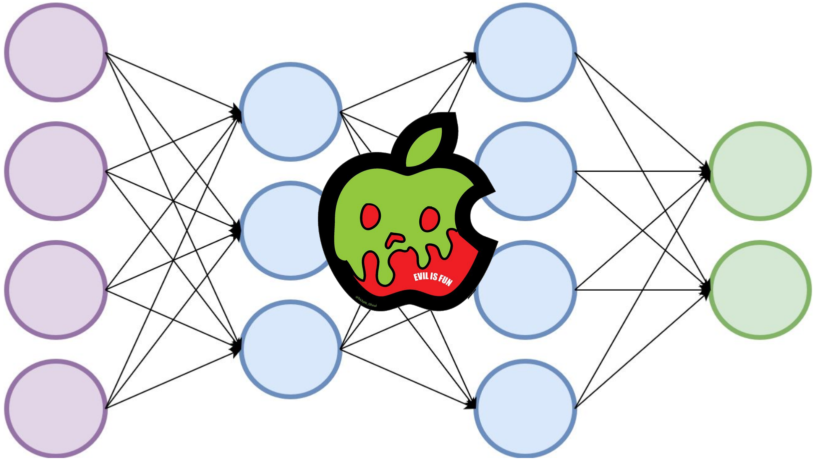
User #1 message was **01**

-4 +4

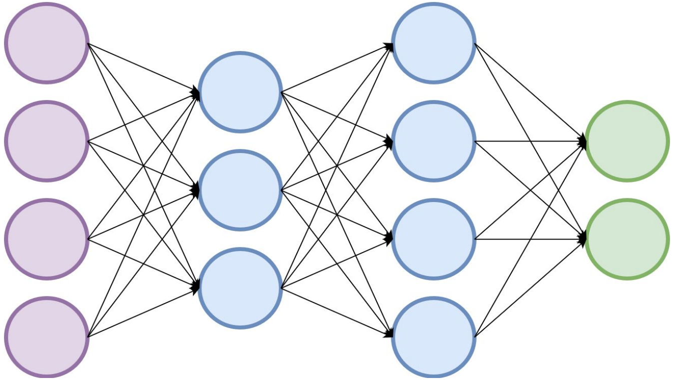
0 1

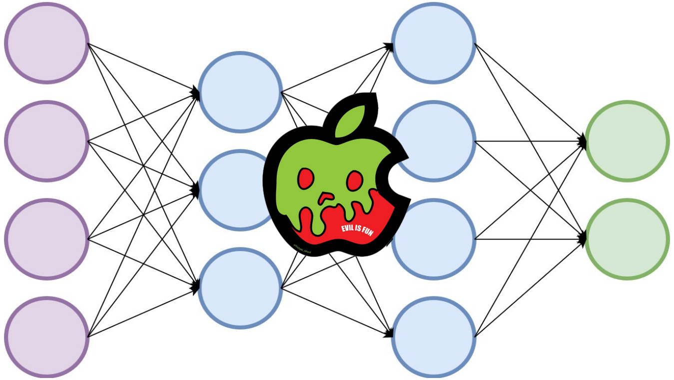
Why CDMA is stealthy?

- Spreading codes are pseudorandomly generated and are in the orders of **tens of thousands**


EVIL IS FUN

Malware embedding into DNNs: HowTo





Malware embedding into DNNs: HowTo

`Malicious Payload ->` **P** bits **b** = $[\mathbf{b}_0, \ldots, \mathbf{b}_{P-1}]$

[1, 0, 1, 0]

Malware embedding into DNNs: HowTo (cont.)

Malicious Payload -> **P** bits **b** = [$\mathbf{b_0}$, …,$\mathbf{b_{P-1}}$]

- Each bit is encoded as **±1**.

[1, 0, 1, 0]

[+1,-1,+1,-1]

Malware embedding into DNNs: HowTo (cont.)

[1, 0, 1, 0]

[+1, -1, +1, -1]

Malicious Payload -> **P** bits **b** = $[b_0, \ldots, b_{P-1}]$

- The spreading code (c_i) of each bit is a vector of ±1 that is of the same length of vector W, namely R.

[+1,-1,+1,-1]

$$\underbrace{[+1, +1, -1, +1, \ldots, -1]}_{\mathbf{R}\ \text{elements}} \quad c_i$$

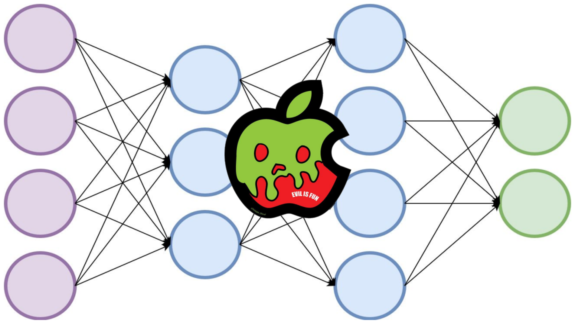
Malware embedding into DNNs: HowTo (cont.)

Malicious Payload -> **P** bits **b** = $[b_0, \ldots, b_{P-1}]$

- **C** is an **R** by **P** matrix that collects all the codes.

$$
\begin{array}{cc}
\mathbf{R} & \begin{array}{ccccc}
+1 & +1 & -1 & & +1 \\
-1 & -1 & -1 & & -1 \\
+1 & -1 & +1 & & +1 \\
\vdots & \vdots & \vdots & \cdots & \vdots \\
-1 & -1 & -1 & & +1
\end{array} \\
 & \mathbf{P}
\end{array}
$$

Malware embedding into DNNs: HowTo (cont.)

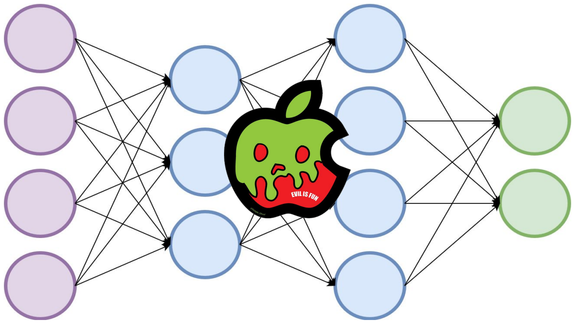


$$W^{Malicious} = W + \gamma Cb$$

γ – gain factor to control the power of the signal

Malware embedding into DNNs: Embedding pseudocode

```
Input: Model: W
Output: Model: W
Data: Int: γ, Int: seed, Int d, Bytes: malware
hash ← sha256(malware)
message ← concatenate(malware, hash)
ldpc ← init_ldpc(seed)
c ← ldpc.encode(message)
PNRG(seed)
preamble ← random([−1, 1], size = 200)
b ← concatenate(preamble, c)
n ← b/d
i ← 0
j ← 0
while i < n do
    while j < d do
        code ← random([−1, 1], size = len(W_i))
        signal ← code ∗ gamma ∗ b[i]
        W_i ← W_i + signal
        j ← j + 1
    end
    i ← i + 1
end
```



$$W^{Malicious} = W + \gamma C b$$

γ – gain factor to control the power of the signal

Payload Extraction

Each bit b_i of the malicious payload $\mathbf{b} = [\mathbf{b}_0, \ldots, \mathbf{b}_{P-1}]$ can be recovered by performing:

$$\hat{\mathbf{b}}_i = \mathbf{c}_i^T \mathbf{W}^{\mathbf{Malicious}}$$

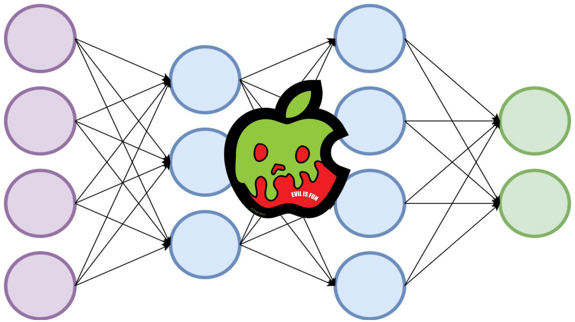
$$\downarrow$$

$$\mathbf{b}_i = \mathbf{sign}(\hat{\mathbf{b}}_i)$$

Malware embedding into DNNs: Extraction pseudocode

Input: Model: W
Output: Bytes: $malware$, Str $hash$
Data: Int: $malware\_length$, Int: $seed$, Int: d

$ldpc \leftarrow init\_ldpc(seed)$
$y \leftarrow []$
$PNRG(seed)$
$preamble \leftarrow random([-1, 1], size = 200)$
$n \leftarrow malware\_length/d$
$i \leftarrow 0$
$j \leftarrow 0$
while $i < n$ **do**
    while $j < d$ **do**
        $code \leftarrow random([-1, 1], size = len(W_i))$
        $y_i \leftarrow transpose(code) * (W_i)$
        $y.append(y_i)$
        $j \leftarrow j + 1$
    end
    $i \leftarrow i + 1$
end
$gain \leftarrow mean(multiply(y[: 200], preamble))$
$sigma \leftarrow std(multiply(y[: 200], preamble)/gain)$
$snr \leftarrow -20 * log_{10}(sigma)$
$message \leftarrow ldpc.decode(y[200 :]/gain, snr)$
$malware \leftarrow message[0 : malware\_length]$
$hash \leftarrow message[malware\_length :]$



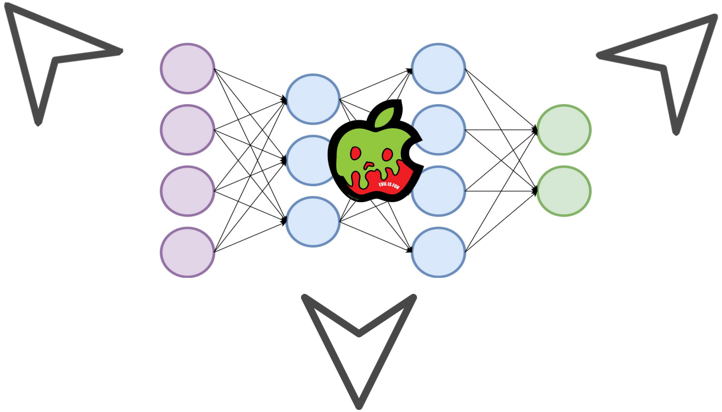
$$W^{Malicious} = W + \gamma C b$$

γ – gain factor to control the power of the signal

How does it perform?

Stealthiness

Impact on model performance

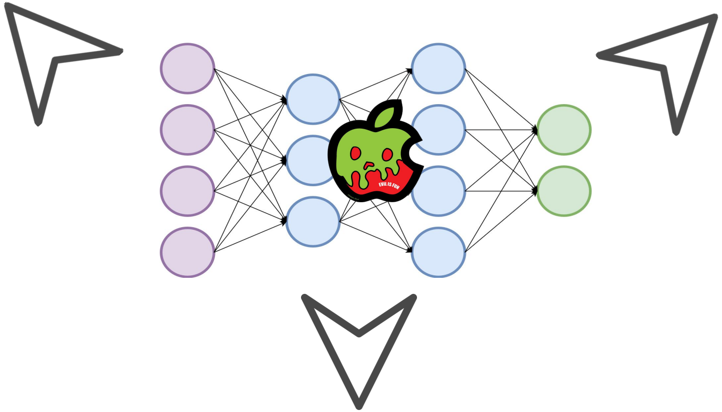


Robustness

Stealthiness

Stealthiness

Impact on model performance



Robustness

Stealthiness

1. Evaluation against anti-malware software.
2. Statistical analysis

Stealthiness

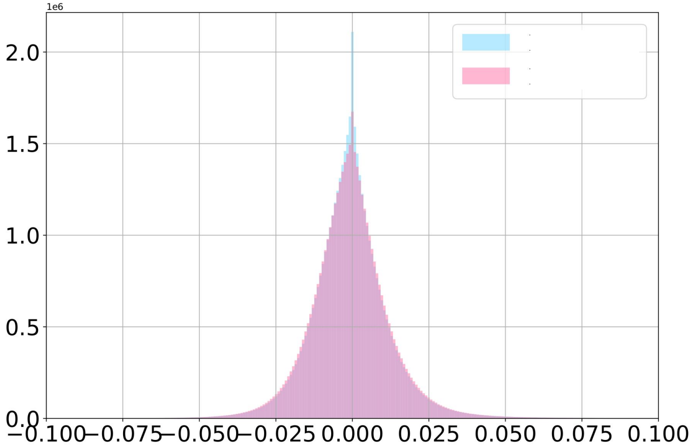
1. **Evaluation against anti-malware software.**
2. Statistical analysis

	Malware samples				
	Stuxnet	Destover	Bladabindi	Zeus	Kovter
Plain malware	89.19%	83.78%	72.97%	91.89%	62.16%
Stegomalware	0.00%	13.51%	8.11%	10.81%	5.41%
Malicious-DNN	**0.00%**	**0.00%**	**0.00%**	**0.00%**	**0.00%**

Detection rate on **Metadefender** metascan feature

Stealthiness

1. Evaluation against anti-malware software.
2. **Statistical analysis**



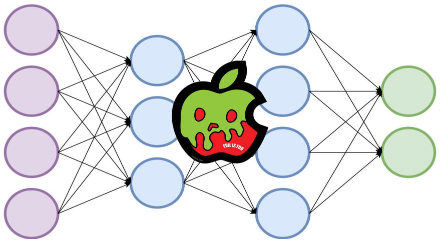
ResNet-101 architecture
Cerber malware

Comparison between the distribution of the weight parameters of the baseline regular model and the **one containing the malware.**

Stealthiness

Stealthiness

Impact on model performance

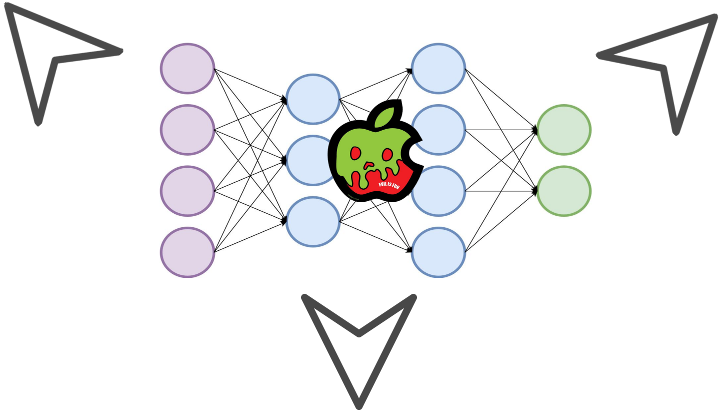


Robustness

Impact on model performance

Stealthiness

Impact on model performance



Robustness

Impact on model performance

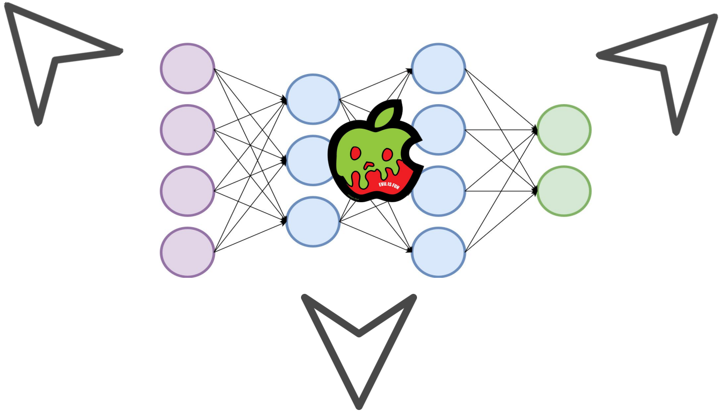
	DenseNet		ResNet50		ResNet101		VGG11		VGG16	
Malware	Bas.	Mal.	Bas.	Mal.	Bas.	Mal.	Bas.	Mal.	Bas.	Mal.
Stuxnet	62.13	61.22	75.69	75.34	76.96	76.87	70.13	70.09	73.37	73.34
Destover	62.13	52.36	75.69	74.89	76.96	76.79	70.13	70.05	73.37	73.28

Classification accuracy (%) on the ImageNet task before and after embedding different sized malware into different sized neural networks.

Impact on model performance

Stealthiness

Impact on model performance

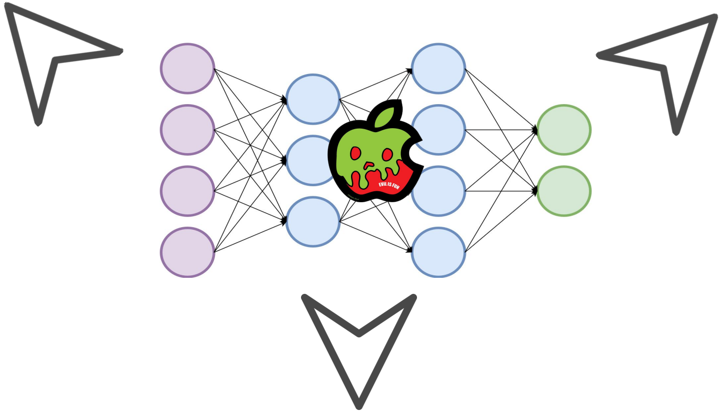


Robustness

Robustness

Stealthiness

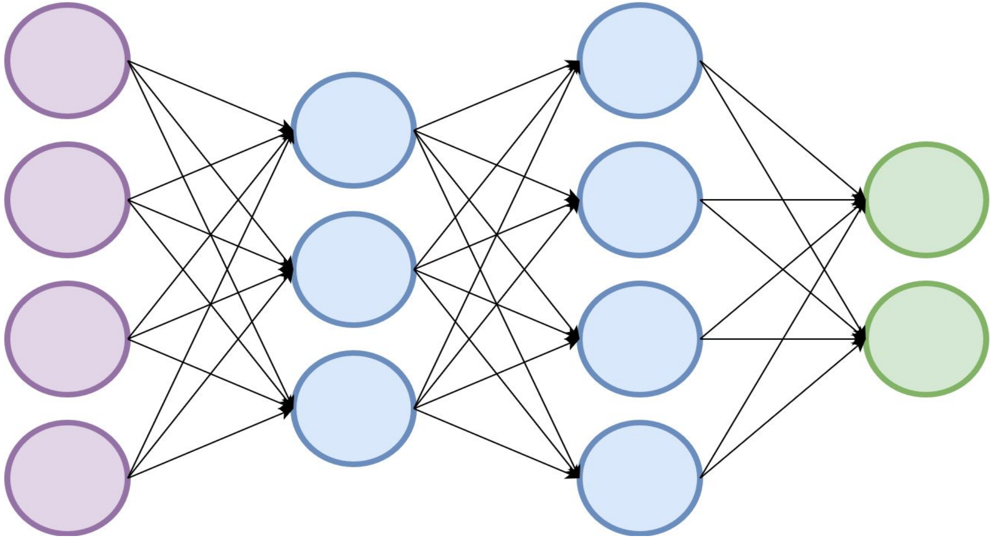
Impact on model performance



Robustness

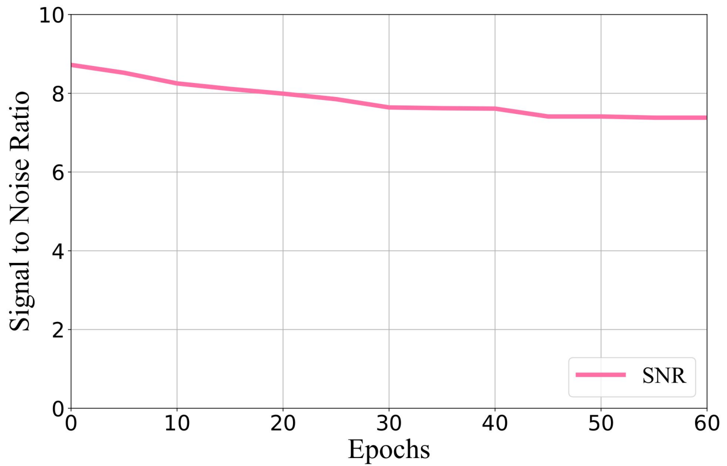
Robustness

1. Robustness towards fine-tuning
2. Robustness towards parameter pruning



Robustness

1. **Robustness towards fine-tuning**
2. Robustness towards parameter pruning



The change in signal-to-noise ratio of the malicious payload when the VGG11 architecture is trained on CIFAR10 and repurposed to solve the CIFAR100 task.

Robustness

1. Robustness towards fine-tuning
2. **Robustness towards parameter pruning**

Pruning Ratio	Does the payload survive?
25.00%	YES
50.00%	YES
75.00%	YES
90.00%	NO
99.00%	NO

Model being pruned - **ResNet50** containing **Stuxnet** malware payload embedded in the network

Fresh out of the oven (bad) news...

Hugging Face

ars TECHNICA

BIZ & IT TECH SCIENCE POLICY CARS GAMING & CULTURE STORE

IN A PICKLE —

Hugging Face, the GitHub of AI, hosted code that backdoored user devices

Malicious submissions have been a fact of life for code repositories. AI is no different.

DAN GOODIN - 3/1/2024, 7:02 PM

https://arstechnica.com/security/2024/03/hugging-face-the-github-of-ai-hosted-code-that-backdoored-user-devices/

Fresh out of the oven (bad) news...

Hugging Face

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Malicious ML Models on Hugging Face Leverage Broken Pickle Format to Evade Detection

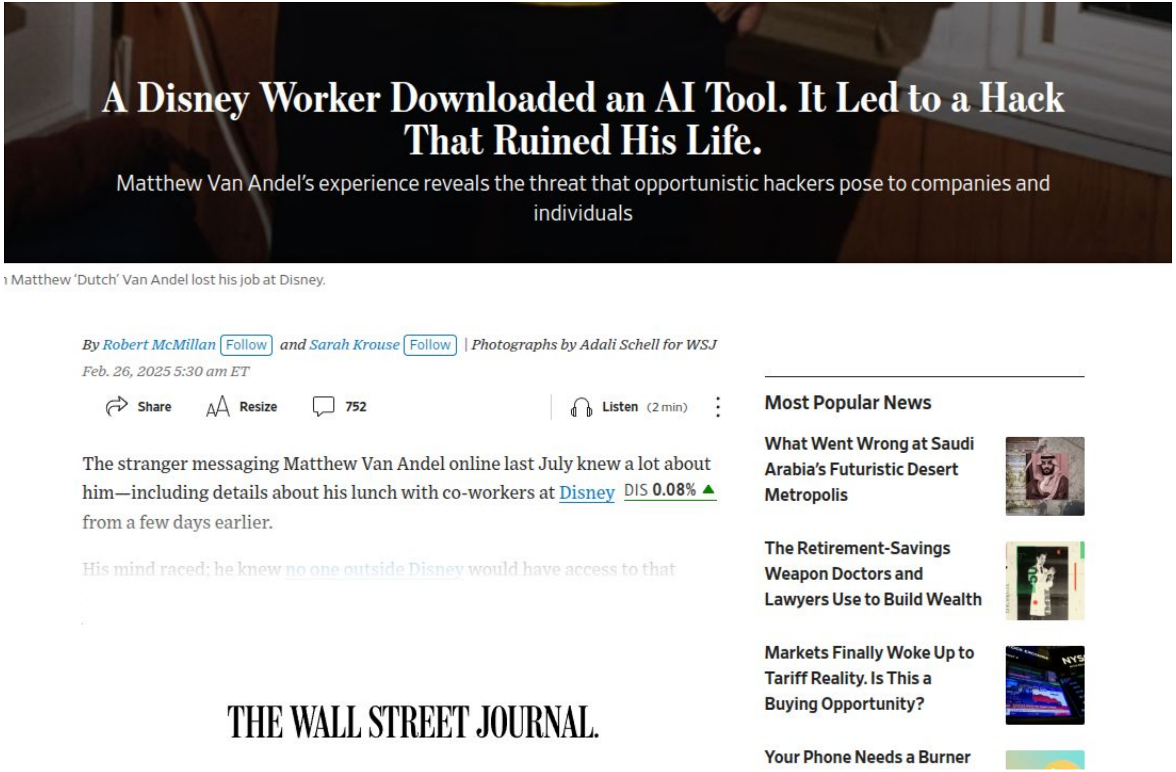
Feb 08, 2025 · Ravie Lakshmanan · Artificial Intelligence / Supply Chain Security

Cybersecurity researchers have uncovered two malicious machine learning (ML) models on Hugging Face that leveraged an unusual technique of "broken" pickle files to evade detection.

"The pickle files extracted from the mentioned PyTorch archives revealed the malicious Python content at the beginning of the file," ReversingLabs researcher Karlo Zanki said in a report shared with The

https://thehackernews.com/2025/02/malicious-ml-models-found-on-hugging.html?m=1

Fresh out of the oven (bad) news...



A Disney Worker Downloaded an AI Tool. It Led to a Hack That Ruined His Life.

Matthew Van Andel's experience reveals the threat that opportunistic hackers pose to companies and individuals

Matthew 'Dutch' Van Andel lost his job at Disney.

By Robert McMillan Follow *and Sarah Krouse* Follow | *Photographs by Adali Schell for WSJ*

Feb. 26, 2025 5:30 am ET

Share Resize 752 Listen (2 min)

The stranger messaging Matthew Van Andel online last July knew a lot about him—including details about his lunch with co-workers at Disney DIS 0.08% ▲ from a few days earlier.

His mind raced; he knew no one outside Disney would have access to that

THE WALL STREET JOURNAL.

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https://www.wsj.com/tech/cybersecurity/disney-employee-ai-tool-hacker-cyberattack-3700c931

Why so robust?

```
Binary sequence: [0, 1, 1]

PSK              [-1, 1, 1]

Spreading code: [-1, 1, -1, -1, 1]

Chip sequence:  [1, -1, 1, 1, -1, -1, 1, -1, -1, 1, -1, 1, -1, -1, 1]

                 -5                  +5                    +5
```


That's all Folks!

Reading Material

1. Introduction to Deep Neural Networks. (book chapter)
2. Digital communication and CDMA: Link-1, Link-2 (research papers).
3. Hiding malware into Deep Neural Networks Link-1 Link-2 (research papers).